

Register-based Data and Statistical Services

By

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Important events in the 1950s

Electronic Computers

- high processing speed
- huge internal storage capability
- processing controlled by internally stored programs

Permanent Population Registers

- **unique and permanent** identification numbers
- **comprehensive and continuously** maintained registers
- **used by central and local public** agencies

Real World System

- Society conceived as different types of **interacting objects**
- Each object type associated with set of **attributes**
- Each object has a **unique** and **permanent identifier**
- At any point of **time**, each object have a set of **attribute values**

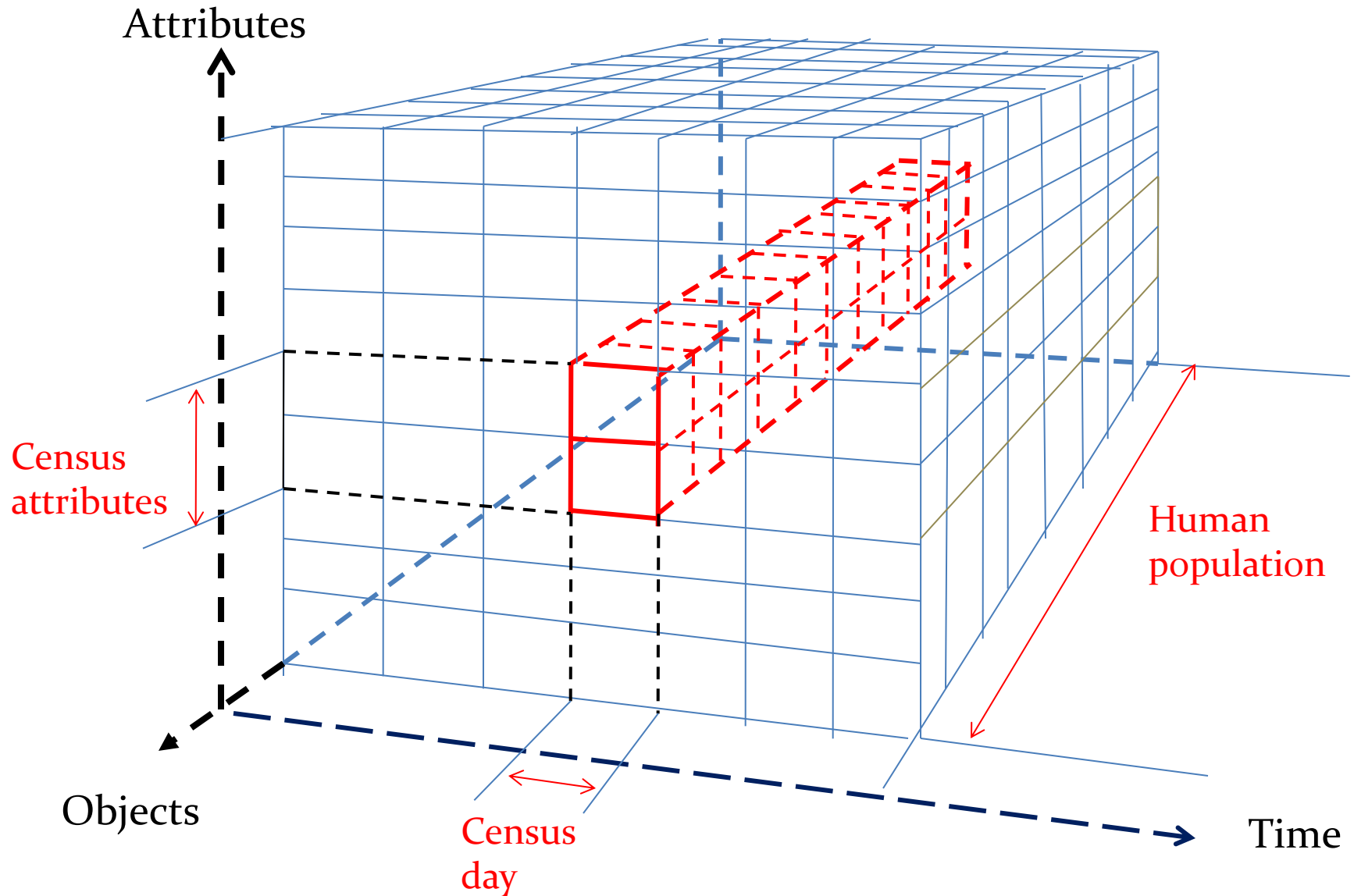
Statistical Register-based/Archive/File Systems

Data collection, storage and use

Basic assumptions:

- Object **registers** with permanent IDs are established, and maintained
- Attribute **catalogues** are established/maintained for all object types
- Data with IDs are **collected** based on the registers
- Data are **organized** and saved by IDs, attributes and time
- Data are **retrieved** on request by IDs, attributes and time
- Data are processed according to **needs** subject to **confidentiality**.

Archive Data Container from the 1960s



Events in Developing Statistical Archive Systems

1960: Paper based on a speech given to the **Meeting of Nordic Statisticians**

1967: **Statistical File System**. Handbook. United Nations.

1965-1970: Papers on **development, uses and legal aspects** of the system

1965: **US National Data Center Proposal** from the Ruggles Committee

1967: Miller, A. R.: **The National Data Center**. The Atlantic

1970-1980: **ARKSY** project in Sweden

1973: Infological Approach to **Data Bases** by Bo Sundgren

1981: First Register-based **Population Census** in Denmark

1990-: More register-based data **applications** and documentation

Selection of Current Applications

- Censuses** (population, location registers)
- Demographics** (demographic registration)
- Foreign trade** (custom data)
- Income** (taxation data)
- Social statistics** (registration of social services)
- Employment** (unemployment registration)
- Education** (registration of students and exams)
- Health** (medical registrations)
- Crime** (police, court, prison registrations)
- Business** (business registers)
- Research** (based on microdata and microdata series)

What about the future?

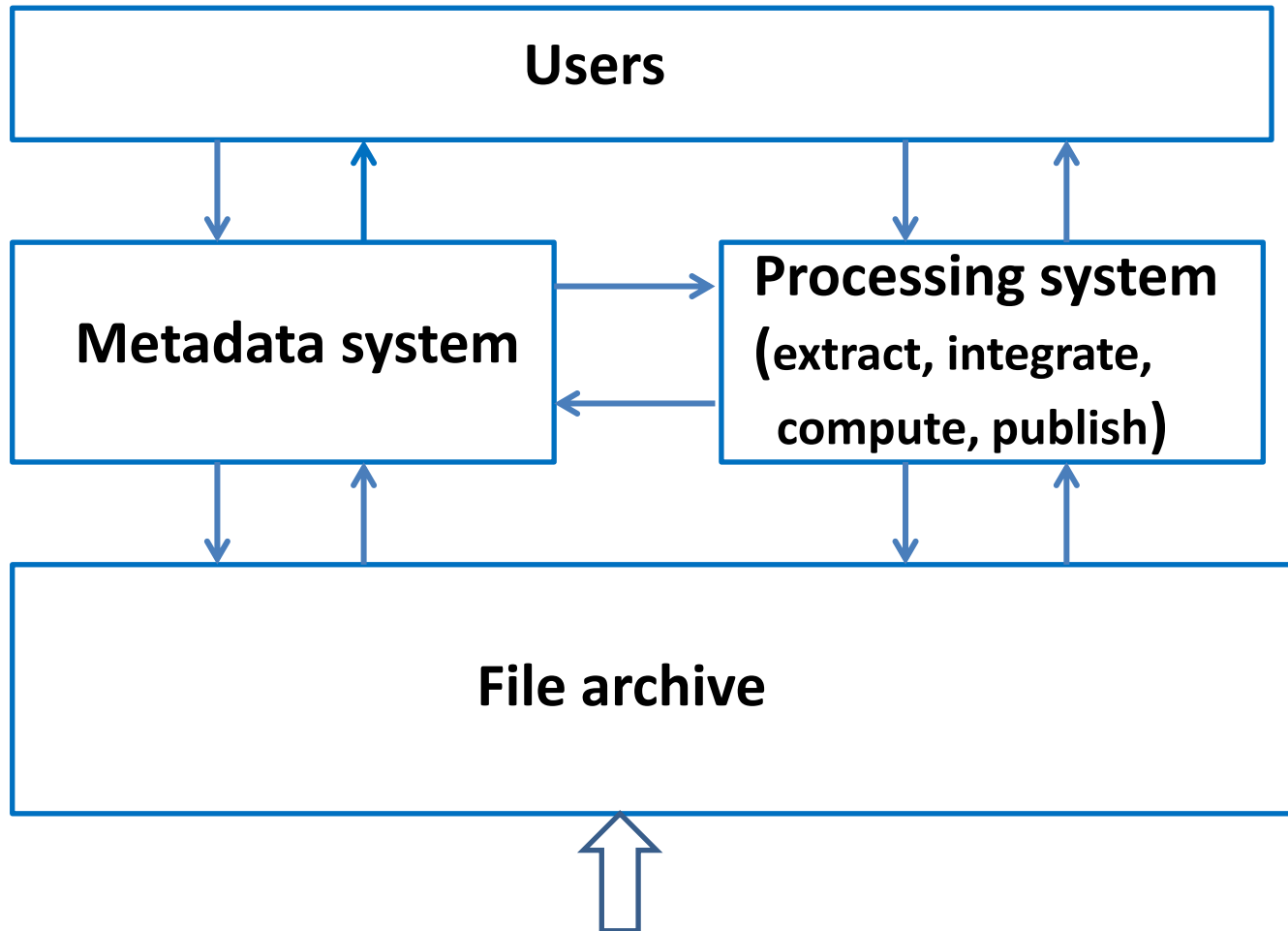
Present Status for Official Statistics

- **Budgets** restrictions for NSIs
- **Response burden** complaints from the public
- Growing **demand** for official statistics
- Continuing **technological** development

Targets for Future Improvement

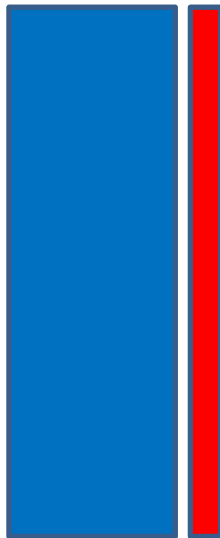
- **Metadata** systems (continuing)
- **Identification** and **integration** (continuing)
- Increased use of **administrative** data (continuing)
- Use of electronic **footprints** (new)
- **Conversion** of **non-numeric** data (new)
- **Storage** and **re-use** of microdata (continuing)
- **Online** and **instant** end user service (continuing)
- **Standardization** and **cooperation** (continuing)

Statistical File System



Administrative and Statistical Data

Volume of units collected



Collection cost/unit



 : Administrative data

 : Statistical data

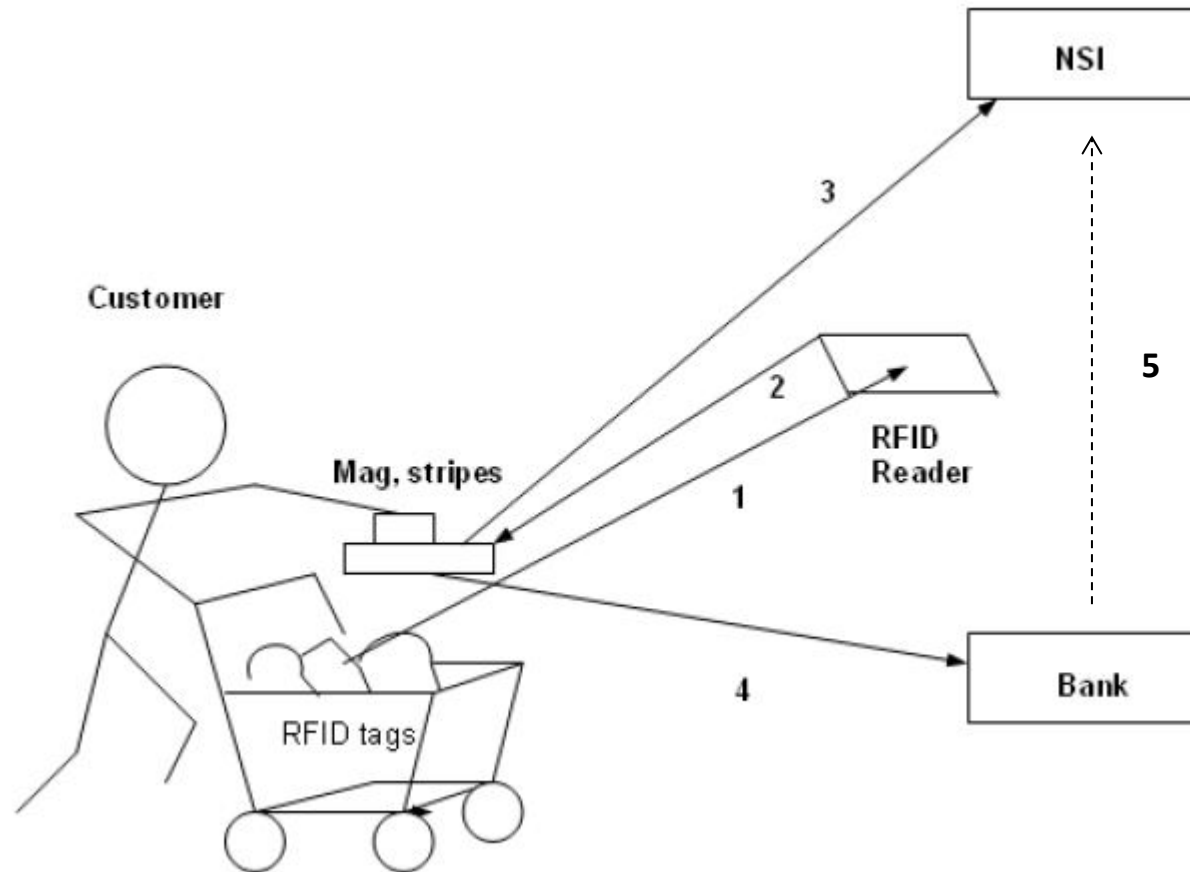
Electronic footprints

Electronic Recording

Examples of electronic recordings which can provide data for official statistics:

1. **Credit card** transactions
2. **Commodity (RFID)** tracking
3. **Toll road (RFID)** recording
4. **Electronic tickets** for travelling
5. **Public services** offered electronically
6. **Immigration** control.
7. **Mobile phone** use
8. **Internet** and social media use
9. **GPS** tracking of traffic and transport
10. **Mixed active/passive** recording

Use of Electronically Observed Data (EOD) in Official Statistics I



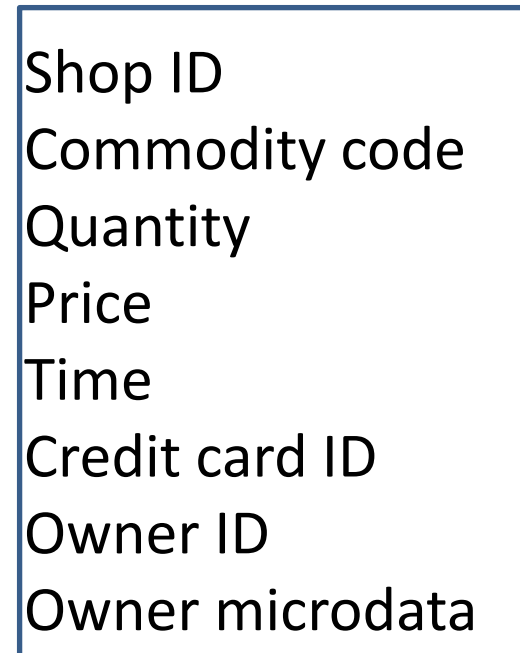
58th World Statistics Congress of the International Statistical Institute, Special Topic Session 25: Analyzing Internet Traffic Flows and digital footprints for statistical purposes, Dublin August 2011.

Shop transaction integrated with NSI microdata

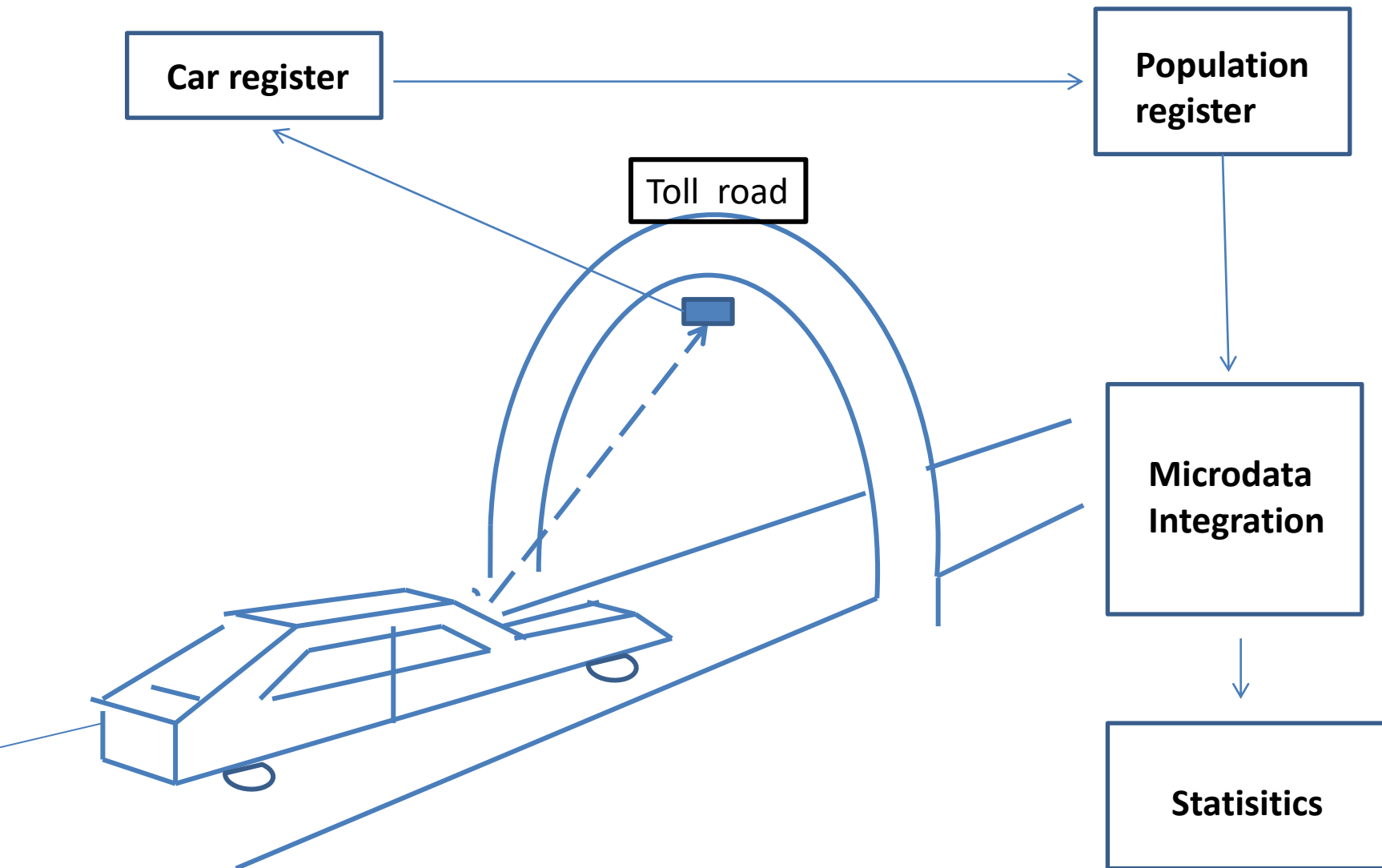
Shop



NSI



Use of EOD in Official Statistics II



Integration of EOD with Existing Data

Recorded EOD

Car number
Location
Time

Car register

Car number
Location
Time
Owner ID

NSI microdata

Car number
Location
Time
Owner ID
Owner microdata

Social media

Some Internet Activities

*Every **minute** of the day:*

100,000 **tweets** are sent

684,478 pieces of content are shared
on **Facebook**

2 million search queries are made on
Google

48 hours of video are uploaded to
YouTube

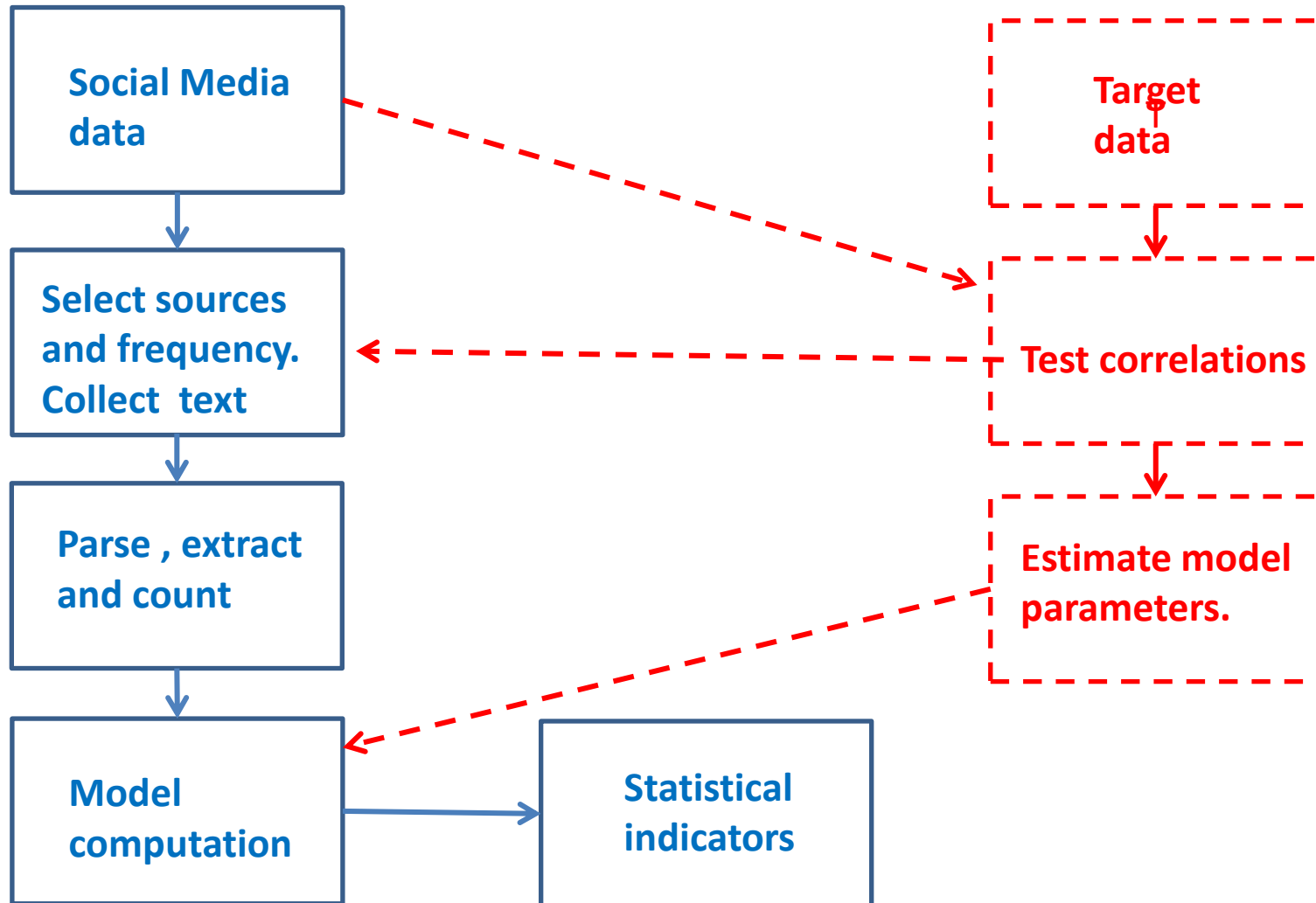
3,600 **photos** are shared on **Instagram**

571 **websites** are created

\$272,000 is **spent** by consumers online

(Source: Wikipedia)

Data for Statistical Indicators



Estimators

$$\text{indicator}_i = f_i (x_{i,j}, x_{i,j+1}, x_{i,j+2}, \dots) \quad i=1, 2, \dots$$

where

indicator is a **target** statistic

$x_{i,j}, x_{i,j+1}, x_{i,j+2}, \dots$ are counted **observations** in social media

f_i is an estimated **relationship** between indicators and observations

End user services

End User Services

- Statistical file (**archive/register-based**) systems
- **Online** and **instant** database service to statistics
- **Remote** electronic access to **microdata**
- **Model-based** statistics
- Pre-approved **analytical** programs
- Access by **mobile** devices

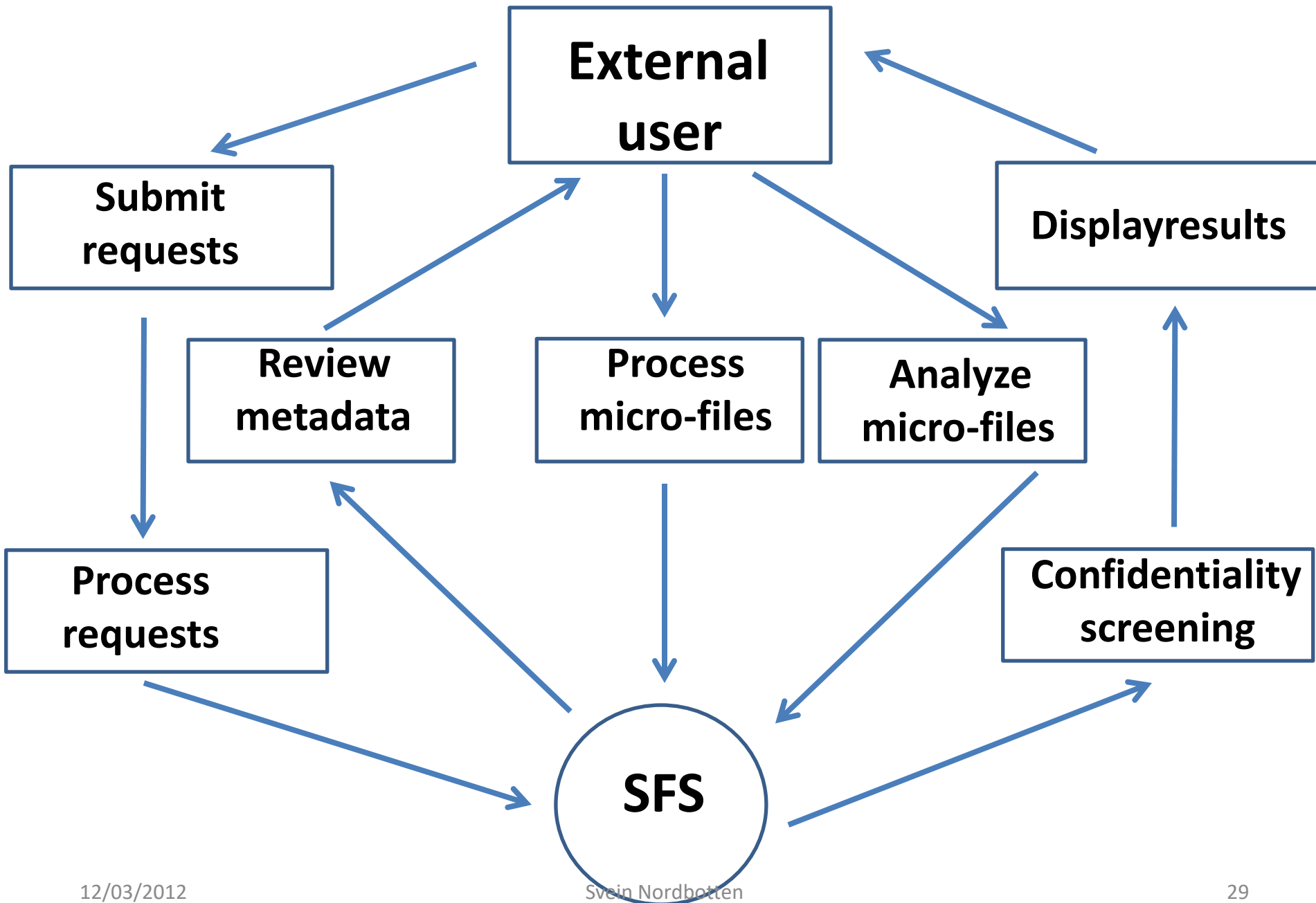
Access and reuse of microdata

Reuse of Microdata

The Confidentiality Dilemma

- Statistical Tables and **Data Cubes**
- **Anonymized** Microdata Files
- Random Error **Distorted** Microdata
- **Remote** Access Facilities
- Data **Laboratories**

UN/ECE (2007): *Managing Statistical Confidentiality & Microdata Access – Principles and Guidelines of Good Practice*. New York and Geneva.



Model-based Statistics and Program Repertoires

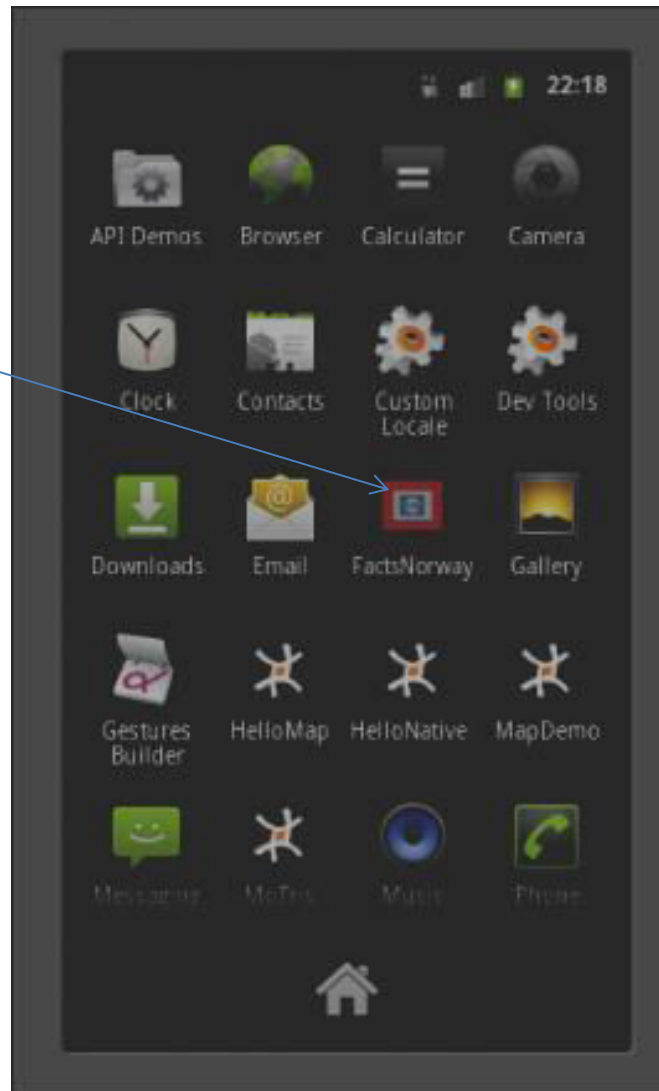
- **Traditional** statistical assumptions
- **Sample survey** assumptions
- Complete surveys regarded as **samples**
- Statistics based **estimates**
- Processing/analytical **program repertoires**

Do users really need to inspect the microdata?

Statistics for everybody

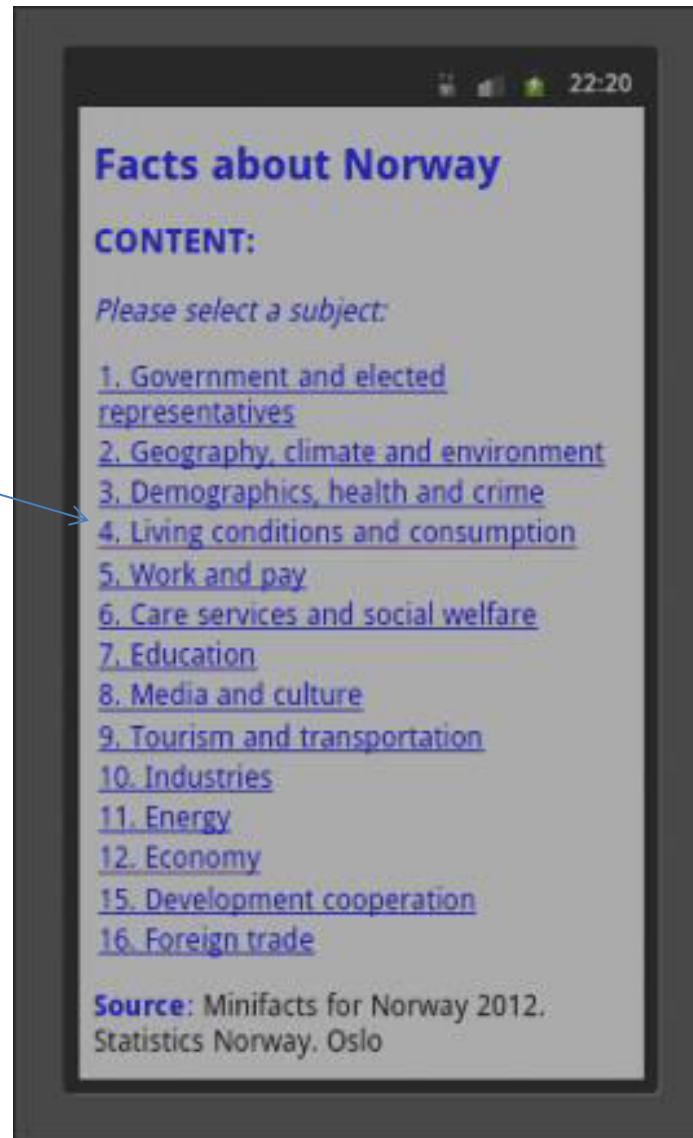
APP for Android smart-phones and devices

FactsNorway



Index to Facts about Norway

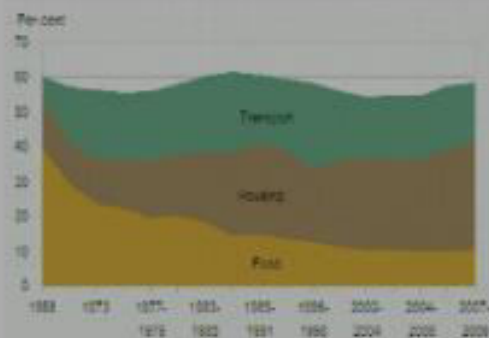
Living conditions
and consumption



4.2 Average annual expenditure per household. 1997-1999 and 2007-2009

Commodity and service category	1999 NOK	Per cent	2009 NOK
Expenditures, total	268 500	100.0	395 100
Food and non-alcoholic beverages	32 400	11.5	46 500
Alcoholic beverages and tobacco	7 600	2.9	10 300
Clothing and footwear	15 200	5.3	20 900
Housing, lighting and heating	69 600	26.7	123 400
Furniture and household articles	16 800	7.0	23 300
Health	6 900	2.8	10 100
Transport	53 300	17.3	64 500
Postal and telecommunication services	5 100	2.5	7 600

4.3 Proportion of household expenditure on food, housing and transport



Source: Statistics Norway.

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Thank you for your attention!

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